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General Outage Probability Model for UAV-to-UAV links in Multi-UAV Networks

Wei Jian Lau [a], Joanne Mun-Yee Lim [a], Chun Yong Chong [b], Nee Shen Ho [c], Thomas Wei Min Ooi [d]

[a] Department of Electrical and Computer Systems Engineering, School of Engineering, Monash University Malaysia
[b] School of IT, Monash University Malaysia
[c] Intel Microelectronics, Ireland
[d] NEX, Intel Microelectronics, Malaysia

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) swarms have become increasingly ubiquitous in the civilian domains due to their maneuverability and scalability in collaborative flying missions. However, there is a lack of characterization of wireless communication performance in multi-UAV system without being constrained by the node spatial distributions. In this paper, the performance of a UAV swarm network is analyzed by characterizing the effects of the UAV-to-UAV (U2U) interference in terms of the Signal-to-Interference-plus-Noise Ratio (SINR). To this end, a closed-form analytical model for the Cumulative Distribution Function (CDF) of the SINR is derived by considering both deterministic and stochastic channel processes, as well as deterministic UAV mobilities. The derived model can be used to analyze U2U interference in a UAV swarm network for any arbitrary node locations. Using this model, the effects of the network parameters on various network Key Performance Indicators (KPIs) such as the outage probabilities and channel capacity rate are studied. The model is extended to incorporate hybrid LoS channels, and the general model is used to evaluate the temporal evolutions of network outage probability given the UAV trajectories. With this, the main advantage of the model in analyzing the baseline outage probabilities of multi-UAV deployments is showcased.

I. Introduction

A. Motivation

UAVs or commonly known as drones, have become ubiquitous and more pervasive in our lives in recent years. These aerial vehicles have been adopted for many civilian use-cases, both recreational or commercial, such as in precision agriculture, remote monitoring, and aerial delivery [1–3].

A multi-UAV system presents many advantages over the single UAV system in terms of cost, flexibility, and fault tolerance, making them well-suited for critical applications [4]. The most popular network architecture for a multi-UAV system is a Flying Ad Hoc Network (FANET) [5], where the main communication links are UAV-to-UAV (U2U) links. In a typical multi-UAV mission, the wireless channel environment changes as the mission progresses due to the 3-Dimensional and dynamic mobilities of these aerial nodes. Proper planning must be carried out prior to the deployment of these UAV swarms so that the UAV trajectory, placement, or transmission power can be optimized for improved baseline reliability of the network [6–8] to ensure successful mission completion. The first step in these optimization efforts is usually to develop an interference model in terms of the SINR. However, the models that are used in these works are generally simple and deterministic models which disregard stochastic fast-fading effects on the wireless signals. As such, this limitation can lead to sub-optimal optimization of the UAV transmission or deployment parameters. Therefore, an accurate characterization on the impact of Co-Channel Interference (CCI) on the U2U performance, considering more realistic channel

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E-mail addresses: wei.lau@monash.edu (W.J. Lau), joanne.lim@monash.edu (J.M.-Y. Lim), chong.chunyong@monash.edu (C.Y. Chong), nee.shen.ho@intel.com (N.S. Ho), thomas.wei.min.ooi@intel.com (T.W.M. Ooi).

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models such as hybrid LoS and stochastic channels, is needed for optimal planning and mission design prior to the deployment of multi-UAV systems in actual environments.

B. Prior Works

The reliability of a UAV communication network under various sources of interferences has been investigated and reported in the literature, each from a unique perspective. Much of these studies focus on characterizing the network reliability and performance in terms of the SINR in the presence of cellular interference [9–13]. Ngo et. al [9] investigated the Ground-to-UAV (G2U) cellular interference for non-fading and Rayleigh fading channels, in a suburban environment. An aerial path loss model is used to model the cellular-to-UAV path loss which is dependent on the elevation angle. Closed-form expressions in terms of the Probability Distribution Function (PDF) of the SINR have also been provided in [10,12] for the downlink transmission from an aerial UAV to a terrestrial cellular Base Station and U2U communications, respectively, with interfering cellular links in the urban environment. The effects of cellular interference by other uplink UAVs and terrestrial base stations on the uplink UAV communications have been studied in [11], from the perspective of outage probability and finite SNR diversity gain. The work by [13] presented analytical models to describe the network outage probability by considering G2U, U2U, and UAV-to-Ground (U2G) interfering links. The U2U co-channel interfering links have also been investigated in [14] by considering directional antennas and in [15] for 3D UAV placements.

A second approach to address this problem reported in the literature is the moment-based approach [16,17]. Using the Moment Generating Function (MGF) together with a novel General Linear Mobility Model, Cong et. al. [16] have presented a method to predict the expected aggregate interference power at the receiver at a future time instance based on the observations in the current time. Similarly, Rached et. al. [17] presented a general moment-based model to describe the SINR for any fading models that can be represented in the form of a power series expansion. However, the CDF-based approach is more popular, albeit more challenging to derive, when analyzing multi-UAV networks because the resultant closed-form CDF can be reused to derive many valuable network KPIs such as the outage probability and channel capacity which is not possible using these MGF-based models.

Most of the CDF models discussed above utilize a common technique known as Stochastic Geometry (SGM) to describe the spatial distribution of the nodes to facilitate the interference modeling and analysis. Poisson Point Process (PPP) [9,14,18] and Binomial Point Process (BPP) [11,12,15] are the two most widely used point processes under the SGM method for deriving an outage probability model. However, both PPP and BPP assume that the nodes are randomly distributed in the given area based on their respective distributions, which is not a practical assumption in a real-life UAV mission. In many real-life UAV missions, the UAVs in a multi-UAV setup have specific formations and set waypoints that can be optimized for navigation to complete the mission efficiently in an area [19–21]. Models derived with this assumption are not suitable for mission-based deployment optimization as only certain stochastic mobility models have been shown to preserve this unique random spatial distribution characteristic such as the Brownian Motion [16,22]. Furthermore, the SGM methods study and characterize the overall network as an ensemble which yields per-network metrics for a link of interest that averages over all possible interfering node locations [23,24]. Despite the compact mathematical formulation that SGM offers, the limitation of this spatial assumption greatly hinders the applicability of these proposed stochastic models in various optimization works where the mobilities of the UAVs represent a target for optimization and cannot conform to the assumption of random distributions.

Besides that, another implication of the 3D aerial mobility of UAVs is the existence of hybrid LoS channels which strongly correlates with the severities of impairments experienced by different links. In the literature, only a few works [6,10,12,13,25] have considered this hybrid channel model for U2U links. In certain multi-UAV formations such as a multi-tiered FANET, as will be discussed in later sections, U2U links can experience Non-Line-of-Sight (NLoS) or Line-of-Sight (LoS) conditions. Therefore, the two main research gaps that are noted from reviewing the existing analytical models are 1) the less accurate interference models for arbitrary UAV spatial models and 2) the less accurate assumptions that are made for the derivation of the models. The aforementioned issues greatly hinder the applicability of existing models in real-world scenarios as the SGM-based models have been shown to underestimate the performance in practical deployments [23].

C. Contributions

To the best of our knowledge, there are limited closed-form CDF models that characterize U2U interference for any arbitrary spatial locations of the UAVs and are based on a practical set of assumptions. To address this gap, we aim to study the effects of U2U CCI on the network performance, specifically the statistical distributions of SINR and Outage Probabilities without the limiting assumption of point process node spatial distributions and considering hybrid LoS channels in this paper. The main contributions of this paper are as follows:

- First, a baseline closed-form expression for the SINR distribution and exact outage probability in the presence of interference from multiple interfering nodes and background noise is derived considering Nakagami fast fading and Friis Propagation Model. No assumptions about the spatial distributions of the nodes are made which allows the model to be evaluated for arbitrary node locations and formations.
- Second, the proposed model is extended by considering hybrid LoS channels for the U2U links. This is modelled as a Bernoulli Process with a U2U LoS Probability model which allows the outage probability to be evaluated in different environments.
- Third, the general outage probability model is used to study the temporal evolution of the main link outage probability for a given set of UAV trajectories obtained from simulations and choice of Modulation and Coding Scheme (MCS). Through this case study, the main contribution of the model in evaluating the expected network performance of a multi-UAV system under hybrid LoS channel and fixed trajectories is demonstrated.

Combining these three contributions, we present the main output of our work which is a closed-form analytical model that characterizes the outage probability of U2U links in multi-UAV networks considering hybrid LoS channels for all active links and deterministic UAV mobility patterns. The proposed model allows the reliability and baseline network performances for mission-based deployments of multi-UAV systems to be studied, which is not possible using existing stochastic outage probability models, as demonstrated in later sections of this paper.

D. Structure

The remainder of the paper is structured as follows. Firstly, we describe the system model, including the Network Architecture, Channel Model, and LoS model used in the analysis of this paper in Section II. We then proceed with Section III that details the derivation of the main outcome of this paper, which is the closed-form mathematical model for the SINR, considering Nakagami fast fading and realistic network parameters. In Section IV, we provide the numerical results obtained by evaluating the closed-form model and compared it with simulation results. A detailed case study is also discussed in this section to analyze the interplay between LoS/NLoS conditions and path loss in urban and suburban environments. Finally, we conclude the paper and discuss the insights that we have gained, along with the significance of the research

in Section V.

II. System model

In this section, we introduce and outline the network architecture, channel model, and fast fading model employed to describe the UAV communications. The notations used throughout this paper are summarized in Table 1.

A. Network Architecture

We consider the deployment of a UAV swarm network, comprising of N transceiving member UAVs and a dedicated UAV-Base Station (UAV-BS). Mathematically, let $\Psi = \{U_1, U_2, \ldots, U_N\}$ denotes the set of member UAVs in the network. In this setup, the UAV-BS acts as a gateway and is responsible for relaying data between the UAV swarm and the Ground Control Station (GCS). To extend the communication range of the FANET, a three-tiered network deployment is considered whereby the UAV-BS is hovering in medium-altitude airspace so that the member UAVs may communication with the GCS. Hence, there exists blockages between the UAV-BS and member UAVs due to the difference in altitude and the probability of blockage is determined by the surrounding environment. In this network, the member UAVs communicate with each other to exchange network and mission information, and with the UAV-BS via U2U links. It is also assumed that the U2U communication links share the same spectrum and hence the network experiences CCI when there are concurrent transmissions from multiple nodes. This architecture is similar to that of a Flying Ad-Hoc Network (FANET) as described in [26]. The UAV-BS will have an additional communication link to the GCS via a UAV-to-Infrastructure (U2I) link, but this is not considered in this paper as the focus is on analyzing the U2U link performance under interference only. The aforementioned architecture is depicted in Fig. 1.

In the following analysis, it is assumed that all the nodes are transmitting Constant Bit Rate (CBR) data to the UAV-BS on the same channel. This assumption is realistic in a real deployment because the UAVs are often required to relay continuous video or image streams in use-cases like remote mapping and precision agriculture [2,3]. From the set Ψ, we consider one member UAV as the main transmitting node which is denoted as U_S. The other nodes, U_{L_u} such that $U_{L_u} \in \Psi \backslash U_S$, constitute the set of interfering UAVs whose transmissions will interfere

TABLE 1
Notations used in the paper and their definitions

Notation	Definition
N	Number of member UAVs
$i \in \{S, L_u\}$	Main link ($i = S$) and Interfering link ($i = L_u$) indices, L_u is the uth interfering link
$e_i \in \{LoS, NLoS\}$	Link conditions for link i
$E = \{e_S, e_{L_1}, \ldots, e_{L_{N-1}}\}$	Set of link conditions for main and interfering links
U_i	The ith member UAV in the network
Ψ	Set of member UAVs in the network
P_i	Transmit power for the link i
h_i	Fast fading power gain for the link i
r_i	Transmitter (Tx) – Receiver (Rx) distance for the link i
d_i^H	Horizontal distance for the link i
d_i^V	Vertical distance for the link i
m_{e_i}	Nakagami-m shape parameter for the link i under condition e_i
θ_{e_i}	Nakagami-m scale parameter for the link i under condition e_i
N_0	Background noise power
α_{e_i}	Path Loss Exponent for the link i under condition e_i
G_{Tx}	Antenna gain at transmitter side
G_{Rx}	Antenna gain at receiver side
$P_O^E(\gamma_{th})$	Outage probability of the main link given the combination of main and interfering link conditions
$P_O(\gamma_{th})$	Unconditioned outage probability of the main link
R	Capacity of the main link

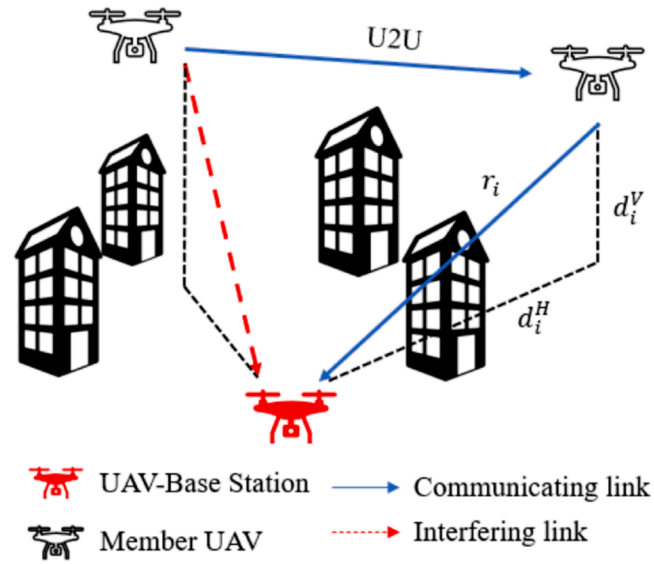

Fig. 1. Illustration of a UAV swarm network architecture deployed in an urban environment. The member UAVs communicate with each other and the UAV-Base Station using U2U communication links. Blue solid lines represent the intended communication link whereas red dashed links represent the unintended interference from the other concurrently transmitting nodes.

with the Signal-of-Interest (SoI) at the UAV-BS. In this topology, the UAV-BS is located at the origin of the 2D horizontal plane of the network area and is hovering at an altitude of z_0 with coordinates $(0,0,z_0)$. When the main transmitter located at (x_S, y_S, z_S) transmits a signal with power P_S to the UAV-BS, the received signal will be interfered by the transmission of the other nodes, U_{L_u}, located at $(x_{L_u}, y_{L_u}, z_{L_u})$ with transmit power P_{L_u}. One way to determine if the main transmission is successful is by analyzing the SINR for the SoI received at the UAV-BS, considering concurrent interference from all other nodes. Since the focus of this paper is on analyzing CCI, interference from other sources is not considered. The interference model is given by:

$$SINR = \frac{P_S \ell(r_S) h_S}{N_0 + \sum_{u=1}^{N-1} P_{L_u} \ell(r_{L_u}) h_{L_u}} \tag{1}$$

(1) represents a standard SINR expression without considering other complex higher physical techniques. In (1), $\ell(r_i)$ represents the path loss as the signal propagates from the main or interfering transmitters to the UAV-BS, $r_i = \sqrt{d_i^{H2} + d_i^{V2}}$, $d_i^H = \sqrt{x_i^2 + y_i^2}$ and $d_i^V = \sqrt{(z_i - z_0)^2}$, is the signal propagation distance in 3D Euclidean space as shown in Fig. 1. and h_i are Random Variables (RV) that model the power gain of the received signal power due to small-scale fading experienced by the signals as a result of multipath effects. For the sake of mathematical derivation and analysis of the U2U SINR model, we consider the link between the main member UAV and UAV-BS to be the main link whereas the other links are considered as interfering links. The same analysis can be generalized to consider other U2U links as the main link, as long as the positions of each UAVs are known.

B. Channel Model

The signals originating from a transmitter at (x_i, y_i, z_i) experiences attenuation when received at the UAV-BS due to the effects of path loss and fading. The path loss function, $\ell(r_i)$, captures both the attenuation from signal propagation loss and antenna gains at the receiver and transmitter side, regardless of the link conditions. The model used for the path loss in this paper is the well-known Friis Propagation Model [27] given by:

$$\ell(r_i) = G_{Tx}G_{Rx}\frac{\lambda^2}{(4\pi)^2 r_i^{\alpha}} \tag{2}$$

where λ is the wavelength of the carrier signal, G_{Tx} and G_{Rx} are the antenna gains of the transmitter and receiver, respectively, and α is the path loss exponent which is dependent on the environment of the network. It is important to note that (2) only highlights the path loss factor from the original Friis Propagation Model as the transmit power is separately denoted in the SINR expression in (1). We assume that all UAVs are equipped with omnidirectional antennas with unitary gain. Hence, $G_{Rx} = G_{Tx} = 1$ and (2) can be represented as:

$$\ell(r_i) = g(\lambda)r_i^{-\alpha} \tag{3}$$

where $g(\lambda) = \frac{\lambda^2}{(4\pi)^2}$ is a constant depending on the choice of wireless access technology for the U2U links. Note that the path loss has an inverse power relationship with the Euclidean distance between the transmitter and receiver.

Furthermore, the signals also experience small-scale fading due to multipath effect which is often modelled using a stochastic model to describe the power gain envelope [28]. The Nakagami-m Fading model is employed in this paper to model the effects of small scale fading due to its general versatility in modelling a wide range of fading conditions and retention of mathematical tractability [15]. Consequently, the RVs in (1) are Gamma Random Variables which quantify the power gain envelope where $h_i \sim \Gamma(m_{e_i}, \theta_{e_i}),\ i\epsilon\{S, L_u\}$ with the Probability Distribution Function (PDF) [28] given as:

$$f_{h_i}(x) = \frac{x^{m_{e_i}-1}}{\Gamma(m_{e_i})\theta_{e_i}^{m_{e_i}}}\exp\left(-\frac{x}{\theta_{e_i}}\right) \tag{4}$$

In (4), m_{e_i} and θ_{e_i} are the Nakagami-m shape and scale parameters for the link i under link condition e_i, respectively, and $\Gamma(\cdot)$ is the Gamma function [29], eq. (8.310.1). These parameters depend on the surrounding environment and different severity of small-scale fading can be modelled from fully NLoS to LoS-dominant conditions by varying these parameters. Due to the probabilistic nature of the fading parameter in (1), the SINR can be characterized as a RV that can be stochastically represented by a certain PDF or CDF.

C. Probability of LoS

The density and height of blockages vary depending on the environment that the UAV swarm is deployed in. This is an important characteristic that must be captured because the blockages directly affect the channel conditions between the member UAVs and UAV-BS. This effect can be modelled by characterizing the Probability of LoS, which is defined as the probability that a transmitter-receiver pair has LoS link in a given environment. The path loss components that are significantly affected are the path loss exponent, Nakagami-m shape, and scale parameters. In this paper, the LoS process between the transmitter and receiver for link i is modelled as a Bernoulli RV, ν_i, with the PDF given by [30]:

$$\mathbb{P}(e_i = LoS) = P_{LoS}\left(z_i, d_i^H\right) = \begin{cases} \left[1 - e^{-\frac{z_i^2}{2c^2}}\right]^{d_i^H\sqrt{ab}}, & z_i - z_0 = 0 \\ \left[1 - \frac{\sqrt{2\pi}c}{d_i^V}\left|Q\left(\frac{z_i}{\gamma}\right) - Q\left(\frac{z_0}{\gamma}\right)\right|\right]^{d_i^H\sqrt{ab}}, & z_i - z_o \neq 0 \end{cases} \tag{5}$$

where $Q(\cdot) = \frac{1}{\sqrt{2\pi}}\int_0^{\infty}\exp\left(-\frac{t^2}{2}\right)dt$ is the Q-function and the area-dependent parameters a, b and c follows the same definition from the ITU-R model [31]. These 3 parameters are used to characterize the buildup of the area of deployment and have the following definitions:

- a – A dimensionless ratio of the area of land covered by buildings to the total land area.
- b – Number of buildings per square area (usually quoted in km^{-2} or m^{-2}).
- c – The scale parameter of the Rayleigh Distribution characterizing the distribution of height of the buildings in the area with the PDF given by:

$$f_{height}(h) = \frac{h}{c^2}e^{-\frac{h^2}{2c^2}} \tag{6}$$

Consequently, the probability of NLoS can be found as:

$$\mathbb{P}(e_i = NLoS) = 1 - \mathbb{P}(e_i = LoS) \tag{7}$$

D. Key Performance Indicators (KPI)

In this paper, the network performance is analyzed using network KPIs that are derived from the SINR distribution. These include the outage probability, coverage probability, and the capacity of the network. In all cases, we firstly define the mathematical definitions for the conditioned KPIs before demonstrating how these conditioned KPIs, especially the conditioned outage probability, can be extended to the general unconditioned case through Probability Theory methods. The conditioned outage probability, $P_o(\gamma_{th}|E)$, is defined as the probability that the SINR of the received signal is less than a certain threshold value given a combination of main and interfering links LoS conditions, E, where the LoS process of each link, e_i, is modelled as a Bernoulli process, i.e. $e_i \in \{LoS, NLoS\}$. The conditioned outage probability is mathematically defined as:

$$P_o\left(\gamma_{th}|E = \left\{e_S, e_{L_1}, \ldots, e_{L_{N-1}}\right\}\right) = \mathbb{P}(SINR\langle\gamma_{th}|E) \tag{8}$$

where γ_{th} is the SINR threshold for successful reception of the signal at the UAV-BS. In other words, the outage probability is defined by the CDF of the SINR conditioned on the set of link conditions E, F_{SINR}^{E}. Therefore, the conditioned outage probability of the SoI can be determined for any combination of U2U link LoS conditions if the conditioned CDF of the SINR is evaluated correctly for the combination.

The capacity of the network can also be derived from the SINR using the Shannon-Hartley Formula:

$$R = B\ log_2(1 + SINR) \tag{9}$$

where B denotes the bandwidth of the main link in Hz. The capacity indicates the maximum transmission rate that can be supported by the channel with an arbitrarily low error rate, which can be useful to both reveal the channel conditions and facilitate the design of an adaptive transmission system to maximize the bandwidth usage. We can also determine the conditioned capacity rate of the channel, defined as the probability that the link can sustain a bandwidth greater than T, given the link conditions, E, from the conditional SINR distribution as:

$$P_R(T|E) = P(R > T|E) = 1 - P_o\left(2^{\frac{T}{B}} - 1\middle|E\right) \tag{10}$$

Note that the expressions in (8) and (10) are conditional on a certain combination of main and interfering link LoS conditions, where E can be any combination from the possible 2^N link conditions. The general and unconditioned expressions, considering all 2^N link LoS condition combinations, can be derived by assuming that the LoS process of each U2U link is modelled as i.i.d RVs and applying the Law of Total Probability.

The assumption of i.i.d holds true in this case as UAVs in a cluster typically maintain a safe distance between one another for safety reasons which means that their placements do not affect the LoS conditions of other UAVs in the network. As the general outage probability model is central to our discussion in later sections, we demonstrate how to obtain its unconditioned expression from (8), considering all 2^N link conditions. By utilizing the models defined in (5) and (7) and the aforementioned Probability Theory method that can be applied due to the i.i.d assumption, the unconditioned model can be mathematically written as:

$$P_o(\gamma_{th}) = \sum_{E=\{e_S,e_{L_1},\ldots,e_{L_{N-1}}\},e_i\in\{LoS,NLoS\}} P_o(\gamma_{th}|E)\Pi_{i\in\{S,L_1,\ldots L_{N-1}\}}\mathbb{P}(e_i) \tag{11}$$

where $\mathbb{P}(e_i)$ is the probability that link i experiences e_i condition, $e_i \in \{LoS,\ NLoS\}$.

III. Performance analysis

In this section, we analyze the performance of the UAV swarm network by characterizing the network KPIs of the main link based on the SINR distribution.

The analysis performed by some studies which omitted background noise or effects from multiple interfering nodes [11,15] may lead to overly optimistic results, especially if the UAVs are flying at similar distances away from one another in the formation. In such a swarm formation, the assumption of only one dominant interfering node does not hold true and the UAV-BS is likely to experience interference from several concurrently transmitting nodes, which further deteriorates the SINR of the main link. Therefore, the performance analysis is carried out by assuming that there are multiple interfering nodes in the network and background noise is non-negligible. Let $I = \sum_{u=1}^{N-1} P_{L_u} h_{L_u} g r_{L_u}^{-\alpha_{e_{L_u}}}$ be the aggregate interference power at the UAV-BS. Note that I constitutes the sum of $(N-1)$ RVs and let x be drawn from I with PDF $f_I(x)$. The CDF of the SINR can then be derived by solving the following integral:

$$F^E_{SINR}(\gamma_{th}) = \int_0^\infty \int_0^{\frac{\gamma_{th}(x+N_0)}{gP_S r_S^{-\alpha_{e_S}}}} f_I(x|E) f_{h_S}(y|E)\,dy\,dx \tag{12}$$

By substituting the PDF of main link fading into (12), the following is obtained:

$$\begin{aligned} F^E_{SINR}(\gamma_{th}) &= \int_0^\infty \int_0^{\frac{\gamma_{th}(x+N_0)}{gP_S r_S^{-\alpha_{e_S}}}} f_I(x|E) \times \frac{y^{m_{e_S}-1}}{\Gamma(m_{e_S})\theta_{e_S}^{m_{e_S}}} \exp\left(-\frac{y}{\theta_{e_S}}\right) dy\,dx \\ &= 1 - \int_0^\infty f_I(x) \int_{\frac{\gamma_{th}(x+N_0)}{gP_S r_S^{-\alpha_{e_S}}}}^\infty \frac{y^{m_{e_S}-1}}{\Gamma(m_{e_S})\theta_{e_S}^{m_{e_S}}} \exp\left(-\frac{y}{\theta_{e_S}}\right) dy\,dx \end{aligned} \tag{13}$$

where m_{e_S} and θ_{e_S} are the Nakagami shape and scale parameters, and α_{e_S} is the path loss exponent for the main link under link condition e_S, respectively. Therefore, these values in (13) are determined by the main link condition, e_S, which the expression is evaluated for. By applying the Complementary CDF (CCDF) of the Gamma distribution and assuming that $m_{e_S} \in Z^+$ and applying [29], eq. (8.354.4), the above can be rewritten as

$$\begin{aligned} F^E_{SINR}(\gamma_{th}) = 1 - \int_0^\infty f_I(x) \sum_{k=0}^{m_{e_S}-1} \frac{1}{k!} \left(\frac{\gamma_{th}}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right)^k \\ \times (x+N_0)^k \exp\left(-\frac{\gamma_{th}(x+N_0)}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right) dx \end{aligned} \tag{14}$$

The derivation of the PDF of RV I is analytically complex because it involves successive convolution of the PDF of the fading RV, h_i. To ensure mathematical tractability, the Laplacian is used to derive the expression. It is immediately recognizable that (14) can be written using the Expectation operator, resulting in:

$$\begin{aligned} F^E_{SINR}(\gamma_{th}) = 1 - E_I\Bigg[\sum_{k=0}^{m_{e_S}-1} \frac{1}{k!} \left(\frac{\gamma_{th}}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right)^k \\ \times (x+N_0)^k \exp\left(-\frac{\gamma_{th}(x+N_0)}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right)\Bigg] \end{aligned} \tag{15}$$

Applying the Binomial expansion on the $(x+N_0)^k$ term reveals:

$$\begin{aligned} F^E_{SINR}(\gamma_{th}) = 1 - \sum_{k=0}^{m_{e_S}-1} \frac{\exp(-A)}{k!} \left(\frac{\gamma_{th}}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right)^k \\ \times \sum_{n=0}^{k} \binom{k}{n} N_0^{k-n} E_I\left[x^n \exp\left(-\frac{\gamma_{th}x}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}\right)\right] \end{aligned} \tag{16}$$

where $A = \frac{\gamma_{th}N_0}{gP_S r_S^{-\alpha_{e_S}}\theta_{e_S}}$. Furthermore, note that $E_I\left[x^n \exp\left(-\frac{\gamma_{th} r_S^{\alpha_{e_S}} x}{gP_S\theta_{e_S}}\right)\right] = \int_0^\infty x^n \exp\left(-\frac{\gamma_{th} r_S^{\alpha_{e_S}} x}{gP_S\theta_{e_S}}\right) f_I(x)dx$ is the n^{th}derivative of the Laplace Transform of the PDF of I multiplied with $(-1)^n$ where $s = \frac{\gamma_{th} r_S^{\alpha_{e_S}}}{gP_S\theta_{e_S}}$. Therefore, the expression can be rewritten as:

$$\begin{aligned} F^E_{SINR}(\gamma_{th}) = 1 - \sum_{k=0}^{m_{e_S}-1} \sum_{n=0}^{k} \binom{k}{n} \frac{exp(-A)}{k!} \\ \times \left(\frac{\gamma_{th} r_S^{\alpha_{e_S}}}{gP_S\theta_{e_S}}\right)^k N_0^{k-n} (-1)^n \left[\frac{\partial^n}{\partial s^n}\mathscr{L}_I(s)\right]_{s=\frac{\gamma r_S^{\alpha_{e_S}}}{gP_S\theta_{e_S}}} \end{aligned} \tag{17}$$

where the Laplacian $\mathscr{L}_I(s)$ is given by:

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} \left(1 + sg\theta_{e_{L_u}} P_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}\right)^{-m_{e_{L_u}}} \tag{18}$$

Proof: See Appendix A

Similarly, $\theta_{e_{L_u}}$ and $m_{e_{L_u}}$ represents the Nakagami shape and scale parameters, and $\alpha_{e_{L_u}}$ is the path loss exponent for interfering link L_u under link condition e_{L_u}. Following from (8), the conditioned outage probability is the CDF of the SINR distribution conditioned on link conditions E as given by (17). Aside from the case of $m_{e_S} = 1$, closed form expressions cannot be obtained readily because it involves computing the nth derivative of the Laplacian. However, the expression can be further simplified and transformed into a closed form expression by applying the General Leibniz Rule [32] to generalize the nth derivative of the Laplacian. Let $(s,n) = \frac{\partial^n}{\partial s^n}\mathscr{L}_I(s)$, the closed form expression for the derivative of the Laplacian is given by:

$$H(s,n) = \sum_{k_{l_1}+k_{l_2}+\ldots+k_{l_{N-1}}=n} \binom{n}{k_{L_1}, k_{L_2}, \ldots, k_{L_{N-1}}} \times \prod_{u=1}^{N-1} (-1)^{k_u} \frac{(m_{e_{L_u}} + k_{e_{L_u}} - 1)!}{(m_{e_{L_u}} - 1)!} \times \left(1 + s\theta_{e_{L_u}} g P_u r_u^{-\alpha_{e_{L_u}}}\right)^{-m_{e_{L_u}} - k_u} \left(\theta_{e_{L_u}} g P_u r_u^{-\alpha_{e_{L_u}}}\right)^{k_u} \tag{19}$$

Proof: See Appendix B

Finally, we present the main contribution of this paper which is a closed-form analytical model that characterizes the exact outage probability of a U2U communication network conditioned on a combination of main and interfering link conditions:

$$P_o(\gamma_{th}|E) = 1 - \sum_{k=0}^{m_{e_S}-1} \sum_{n=0}^{k} \binom{k}{n} \frac{\exp(-A)}{k!} \left(\frac{\gamma_{th}}{g P_S r_S^{-\alpha_{e_S}} \theta_{e_S}}\right)^k (-1)^n N_0^{k-n} \times \sum_{k_{L_1}+k_{L_2}+\ldots+k_{L_{N-1}}=n} \binom{n}{k_{L_1}, k_{L_2}, \ldots, k_{L_{N-1}}} \times \prod_{u \in \Psi \setminus U_S} (-1)^{k_u} \frac{(m_{e_{L_u}} + k_u - 1)!}{(m_{e_{L_u}} - 1)!} \left(1 + s\theta_{e_{L_u}} g P_u r_u^{-\alpha_{e_{L_u}}}\right)^{-m_{e_{L_u}} - k_u} \times \left(\theta_{e_{L_u}} g P_u r_u^{-\alpha_{e_{L_u}}}\right)^{k_u} \tag{20}$$

By combining (20) with (11), it is now possible to evaluate the outage probability of the multi-UAV network considering both hybrid LoS channels and deterministic UAV trajectories.

IV. Numerical results

In this section, we present two types of numerical results. Firstly, we compare the numerical results obtained by evaluating the closed-form expressions derived in the previous section to the simulation results under different operational network conditions to verify the accuracies of the model described in (20). To obtain the simulation results for the SINR and outage probability, Monte Carlo simulations with 4000 network realizations are performed using MATLAB. To further showcase the practicality of the proposed mode, a benchmarking evaluation is performed with one stochastic and one deterministic model via Monte Carlo simulations. Next, we demonstrate how the model can be used to evaluate the temporal outage probability of the network given the UAV trajectories obtained from a mobility trace which showcases the flexibility of our proposed model due to the exclusion of a spatial distribution assumption. Table 2 tabulates the fixed parameters that are used for evaluating the proposed model and running the simulations to get the results for the first three subsections. Subsections IV.A and Subsection IV.B presents the outage probabilities of the network evaluated for different severities of path loss and fading, respectively. In Subsection IV.C, we discuss the channel capacity of the network derived from the proposed model, considering different severities of fading and path loss. In Subsection IV.D, results for benchmarking the proposed model against existing stochastic and deterministic models are presented. Finally in Subsection IV.E, the main strength of the model is demonstrated by evaluating the outage probability of a multi-UAV system with defined trajectories in urban and suburban environments. In the subsequent subsections, 4dB is used as the SINR threshold for an outage event, unless otherwise specified.

TABLE 2
Fixed Parameters

Parameter	Value	Parameter	Value
P_S	10 mW	N_0	-86 dBm
P_{L_u}	10 mW	$\theta_{e_{L_u}}$	$\frac{1}{m_{e_{L_u}}}$
G_{Rx}	0 dB	θ_{e_S}	$\frac{1}{m_{e_S}}$
G_{Tx}	0 dB	f	2 GHz
N	4	r_s	75m
$r_{L_{\{1,2,3\}}}$	100m		

A. Severity of Path Loss

The severity of the path loss can be captured by the value of the path loss exponent, α. In this case, a LoS channel is modelled with a path loss exponent of 2, indicating free space path loss. On the other hand, path loss exponent of 3.5 is used to model the path loss of a NLoS channel, in an urban environment [33]. Fig. 2(a) plots the network outage probability for 2 different severities of path loss. In this case, the links are assumed to undergo attenuation with the same path loss exponent, $\alpha = \alpha_S = \alpha_{L_u}$, so that the effects of the path loss exponent on the outage probability can be compared. Due to this assumption, the conditioned model in (20) is evaluated. To obtain these results, it is assumed that $m = m_s = m_{L_u} = 4$. The plot in Fig. 2(a) shows that the analytical model closely matches the simulation results, which verifies the accuracy of the model in different path loss conditions. From Fig. 2(a), it can be observed that the trend of the outage probability is similar in both simulated environments with the major difference being the point of absolute outage in the network, wherein after the outage probability approaches one as the SINR threshold increases. In this case, increasing α leads to higher path loss exponent which indicates a shift from LoS condition to NLoS condition in the simulated environment. Hence, a higher path loss exponent results in the signals experiencing higher attenuation losses which lowers the main received signal power at the receiver. Although the aggregate interference power is similarly attenuated, the change in the received power of the main signal is more significant in this case due to its closer proximity to the UAV-BS. With lower received power, the resultant SINR is lower as well which leads to a higher outage probability at any given SINR threshold before the point of absolute outage compared to the environment with lower α value. A higher path loss exponent will result in a lower threshold for absolute outage compared to the converse case, thereby making the communication links less tolerant towards external noises and interferences, as well as the placement of the UAVs in the cluster. This observation indicates that the path loss exponent has a significant impact on the outage probability of the network and can affect how the multi-UAV network is optimally deployed to work around the high outage probabilities in certain environments. From these observations, it can be concluded that the path loss exponent has a direct effect on the outage probability because the Friis Propagation Model used in the derivation is a deterministic model.

B. Severity of Fading

As mentioned previously, the severity of fading that the signals experience can be modelled using the m parameter of the Gamma distribution. The severity of the fading conditions has an inverse relationship with the value of the m parameter. Note that when $m = 1$, the Nakagami-m fading reduces to Rayleigh Fading which implies a fully NLoS channel condition between the Tx and Rx whereas higher m values such as $m = 4$ can sufficiently model a LoS channel condition [34]. It is assumed that $m = m_{L_u} = m_S$. The path loss exponents, α, of all links are set to 2 to represent Free Space Path Loss and the Gamma scale parameters, θ_S and θ_{L_u}, are set as the reciprocal of the m parameter of the respective links so that the mean of the fading power envelope is 1. Fig. 2 (b) presents the plot of the outage probability of the main link obtained using the proposed model and simulations for different values of m.

The plot shows that the analytical results obtained via evaluation of the proposed model agrees with the outage probability calculated from the Monte Carlo simulations with varying severities of fading. It can be observed from the plot that the effect of fading on the outage probability of the multi-UAV network is not as straightforward to determine as the change in severities of path loss. This difference surfaces due to the stochastic nature of the Nakagami-m fading model as compared to the

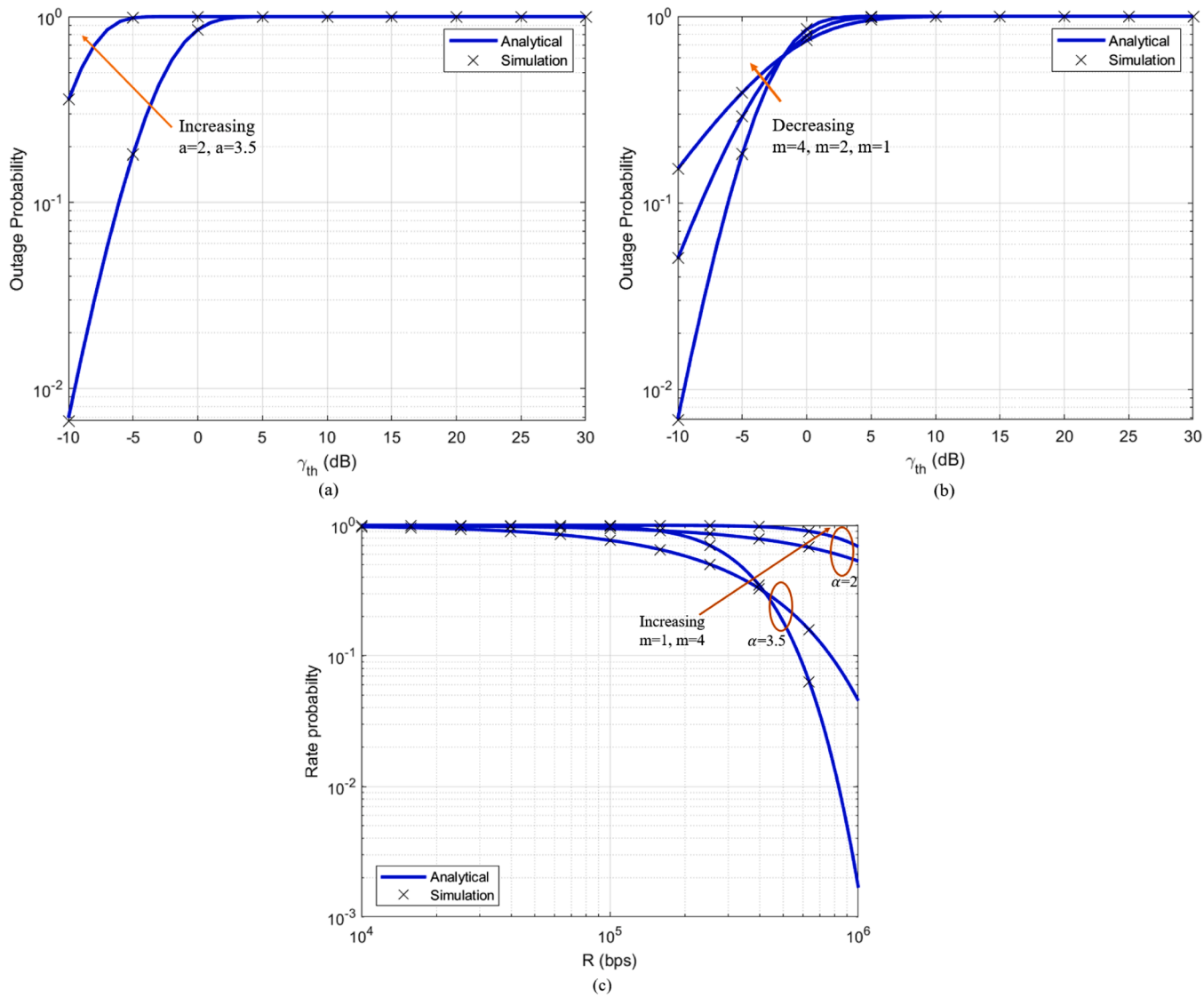

Fig. 2. Outage probability of the main link for different severities of (a) path loss (b) fading. (c) Channel Capacity rate under different network conditions

deterministic nature of the Friis Propagation Model. Interestingly, it can be observed that there is a crossover point present in the plot whereby the behavior of the outage probabilities of different fading severities changes. This observation can be explained by the relationship between the fading severity and variance of the received power whereby less severe fading results in lower variance of the received power which leads to a lower probability for the SINR to be higher than the threshold at high SINR threshold regimes. Nevertheless, the proposed model can be used to evaluate the outage probability of a multi-UAV network when deployed in different environments with varying levels of obstruction which affect the severity of fading experienced by the channels.

C. Capacity Analysis

In this subsection, we characterize and analyze the network performance in terms of the channel capacity. This analysis is particularly insightful in revealing the probability that the channel can accommodate at least a certain transmission bitrate with an arbitrarily low error rate. Using (10), we can evaluate the Complementary CDF (CCDF) of the channel capacity for different network parameters. Fig. 2(c) presents the CCDF of the channel capacity, termed rate probability, evaluated for different networks with different combinations of path loss and fading severities. It can be observed that as the channel becomes more impaired, either with an increase in severity of path loss or fading, the channel capacity rate decreases. This is because as the signals suffer more attenuation losses, either from path loss or fast fading, the SINR decreases, and they become less reliable with higher outage probability. As such, the probability for the link to sustain a high channel capacity decreases due to this reduction in reliability. Another observation that can be made is that for the same path loss, α, the channel capacity rate of the main link decreases at a faster rate for the more favorable fading conditions ($m = 4$) compared to the condition with more significant fading effects ($m = 1$). This behavior leads to the existence of the crossover point, wherein after the probability for the links to sustain higher capacity is higher in the environment with more significant fading due to higher variance in the received power as dictated by the Nakagami-m model which causes a slower drop-off.

To summarize, this subsection has demonstrated the advantage of deriving an exact CDF model for the SINR compared to deriving it as an MGF-based model as it can be adopted to study different network KPIs such as the channel capacity rate under different severities of path loss and fading. This analysis can potentially be used in the planning phase to design and optimize the UAV transmission systems so that the channel can sustain a certain load with a predefined confidence level. After verifying the accuracy of the proposed model in these three subsections, the model will be used in the subsequent subsections on the

benchmarking and case study.

D. Benchmarking

In this subsection, we compare the accuracy of the proposed model with a closed-form stochastic outage probability model proposed by Armeniakos, Bithas and Kanatas [15] and a deterministic model in a simulated multi-UAV network. The benchmark stochastic model is chosen because it shares similar major network assumptions with those utilized in derivation of the proposed model such as the channel models used, i.e., Friis Propagation Loss and Nakagami-m Fast Fading models, assumption of a half-duplex communication system and characterization of U2U link outage probability only. However, two major differences in assumptions exist between the benchmark and proposed model which are the stochastic node spatial distributions and assumption of only one dominant interfering node for the benchmark model. The first difference in terms of spatial distribution can be mitigated by performing a second Monte Carlo simulations of the interfering node spatial locations using the PPP distance distribution conditioned on a fixed main link distance, r_S, on top of the usual Monte Carlo simulations of network realizations when performing this benchmarking exercise. With this extra layer of Monte Carlo simulations, the position of each interfering UAV is sampled from the PPP distance distribution and used for evaluations of outage probability using the proposed model, effectively making the UAV positions deterministic. As per the PPP distance distribution, it is also assumed that all nodes exist within a sphere of fixed radius, R.

Furthermore, we also compare the accuracies of both models with a deterministic outage probability model that is used for multi-UAV trajectory optimization in [7]. This deterministic model does not consider stochastic channel effects such as fast fading which is a common practice in prior works that have used the SINR expression in multi-UAV system deployment optimizations. As such, this benchmarking exercise serves to demonstrate how the proposed model performs against existing stochastic models with limiting network assumption (only one dominant interfering node) and deterministic models that are commonly used in optimization works. The parameters that are used for this benchmarking exercise are tabulated in Table 3.

Fig. 3 plots the outage probabilities obtained from the evaluations of our proposed model, the benchmark stochastic model [15] and the deterministic model [7] compared with the outage probabilities from Monte Carlo simulations. As observed from the plot, the outage probabilities obtained from the proposed model agrees with the probabilities calculated from Monte Carlo simulations. On the other hand, it is observed that the benchmark stochastic model severely underestimates the Monte Carlo outage probabilities because of its assumption of only one dominant interfering node. As such, the benchmark model does not consider effects from the other interfering nodes other than the closest one to the receiver. From this exercise, it can be concluded that this assumption is not suitable for a multi-UAV network as it oversimplifies the actual network and causes significant underestimation when the number of nodes in the network increases. Conversely, it is observed that the deterministic model overestimates the actual outage probabilities and provides a false insight that the network reaches absolute outage at a much lower SINR threshold. This is because the absence of fast fading effect results in lower attenuation for all signals in the network, including the interfering signals. As the interfering signals experience less attenuation, the aggregated interference power at the receiver increases which subsequently leads to lower SINR for the main signal and higher outage probability at any given SINR threshold.

TABLE 3
Parameters for benchmarking simulations

Parameter	Value	Parameter	Value
$P_S = P_{Lu}$	1 mW	Simulation Area	R=100m
$G_{Rx} = G_{Tx}$	0 dB	N_0	0 dB
N	10	$m_{L_u} = m_S$	3
r_S	20m	$\theta_{L_u} = \theta_S$	1
$\alpha_{L_u} = \alpha_S$	2		

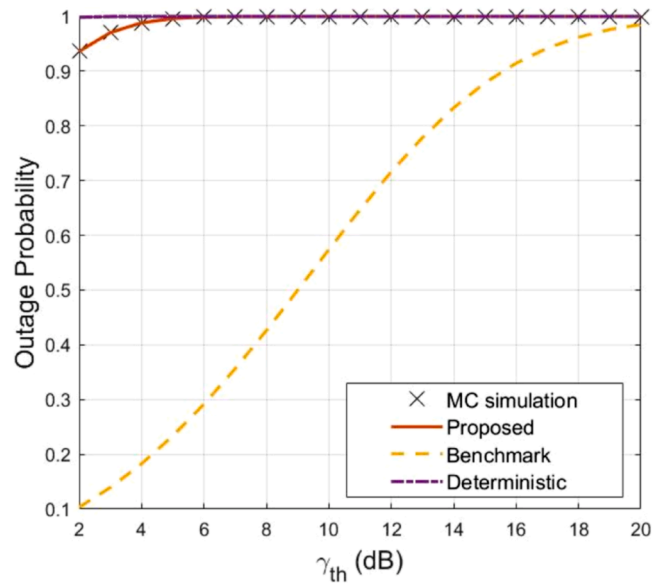

Fig. 3. Outage probability with different SINR thresholds for the Monte Carlo simulations, our proposed model, benchmark stochastic and deterministic models

To summarize, this benchmarking exercise has demonstrated the significance of our proposed model compared to other existing models. Existing stochastic models severely underestimate the actual outage probabilities due to their limiting assumptions whereas the deterministic model overestimates the actual outage probabilities due to the omission of stochastic fast fading effects that actually attenuates the interfering signals in favor of the SINR of the main signal.

E. Temporal Outage Probability

In this subsection, we demonstrate how to integrate the proposed model in (20) with (11) to evaluate the general outage probability of the main link in the multi-UAV network. The main advantage of the model

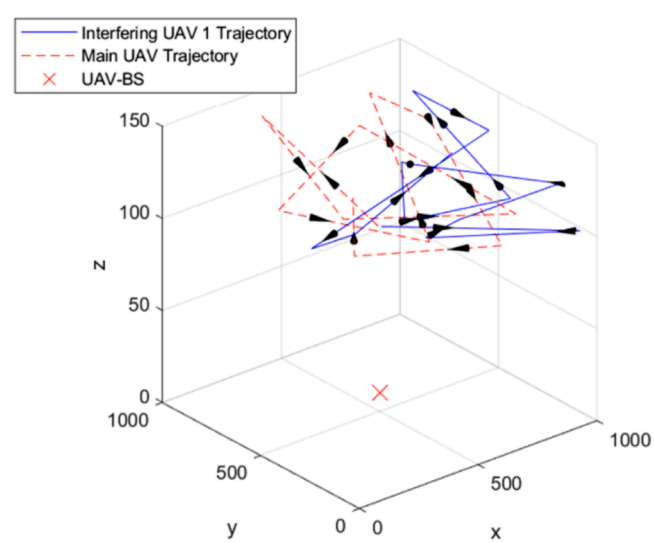

Fig. 4. Trajectories of the main UAV and one of the interfering UAV in the mission

in (20) is that it does not make any assumptions about the distributions of the UAVs which allows the trend of the outage probability to be investigated with time for a given set of UAV trajectories. This was previously not possible using the existing stochastic SINR models in the literature due to the Point Process assumptions. For this demonstration, we use mobility trace data obtained from OMNeT++ simulations [35] which is visualized for the main and one interfering UAV in Fig. 4.

To study the outage probability of the network under hybrid LoS conditions in different deployment environments, we adopt the values of the building parameters from [36] to be used with the LoS probability model in (5), where $(a,b,c) = (0.3,500,15)$ for urban environments and (0.1,750,8) for suburban environments. The parameters used for the simulation are tabulated in Table 4. In the simulation, 5 UAVs move with Random Waypoint Mobility Model (RWPM) in the Area of Deployment while communicating with a static UAV-BS located at the center of the mission space. The position of each of the UAV is recorded in 0.5 seconds interval to form the mobility trace data.

At each timestep of the simulation, the position of each UAV is obtained from the mobility trace and used to evaluate the outage probability of the main link using the combination of the proposed model and the LoS probability model, as represented in (11). As such, the outage probability of the multi-UAV system can be evaluated in different environments given the same mobility trace which is an imperative feature in the planning phase of a real-life UAV mission. As the model needs to be evaluated for all possible 2^N combinations of link conditions, we present how this can be implemented in MATLAB in Algorithm 1 with complexity $\mathscr{O}(2^N * N)$.

The inputs to Algorithm 1 are the positions of each UAV i at time t as extracted from the mobility trace data. It firstly initializes an indicator matrix, $\boldsymbol{J} \in \mathbb{R}^{2^N \times N}$, to store all possible combinations of main and interfering link conditions. Therefore, each row of $\boldsymbol{J}$ represents one combination, $E = \{e_S, e_{L_1}, \ldots, e_{L_{N-1}}\}$. Thereafter, the algorithm iterates through each individual element of all rows of $\boldsymbol{J}$ and calculate the resultant product of the LoS probability term for each set of E, $P_{res} = \Pi_{c=1}^{N} \mathbb{P}(e_c)$ as seen in Step 11 or 14 of Algorithm 1. At the same time, steps 10 and 13 stores the corresponding Nakagami shape and scale parameters, and the path loss exponent corresponding to link condition e_c in the vectors $\boldsymbol{M}$, $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$. After iterating through all links and collecting the required LoS-dependent parameters as well as the resultant probability terms, the vectors containing the LoS-dependent parameters are passed to (20) in step 17 to obtain the resultant conditioned outage probability for the current set of link LoS conditions, E. After the outer loop completes, the final general outage probability for time step t, $P_O(t)$, is computed through matrix multiplication of $\boldsymbol{P}_{res}^T$ and $\boldsymbol{P}_O$.

The outage probability evaluated using the proposed model and Algorithm 1 is plotted in Fig. 5 against the mission time for the recorded mobility trace data in urban and suburban environments. As observed from Fig. 5, the main link in the FANET is likely to experience outage events with high probability throughout the mission period for the given UAV trajectories. This implies that the UAV trajectory and placement is not optimally designed for the mission in the deployment environment as the high Outage Probabilities imply that the UAV-BS is not able to reliably decode the SoI throughout the mission. Throughout the mission period, the main link is also more likely to experience outage when deployed in the suburban environment compared to the urban environment with the same UAV trajectories. This is because LoS link conditions are more probable in a suburban environment compared to an urban environment. As such, the links are more likely to experience less severe path loss and fading in the suburban environment due to the prominence of LoS conditions as dictated by the LoS model presented in (5). In this case, both the main and interfering links are subjected to lower attenuation losses which result in higher received signal and aggregate interference powers at the receiver side. Therefore, the increase in aggregate interference power overwhelms the increase in the received power of the SoI which results in lower overall SINR for the SoI in the suburban environment compared to the urban environment. This drop in SINR results in the observation of higher outage probability for the SoI as the links become less reliable in the suburban environment.

Fig. 6 visualizes the position of the main and interfering UAVs in the area of deployment during periods of low ($t = 200s$) and high ($t = 500s$) outage probabilities. As observed, the outage probability of the

TABLE 4
Parameters for simulations

Parameter	Value
$\boldsymbol{P_S} = \boldsymbol{P_L}$	10 mW
$\boldsymbol{G_{Rx}} = \boldsymbol{G_{Tx}}$	0 dB
$\boldsymbol{N}$	5
Simulation Area	1km x 1km x 150m
UAV-BS coordinates	(500m, 500m, 10m)
UAVs altitude range	100m – 150m
Speed	10m/s
Duration	10mins

Algorithm 1
Evaluation of the general Outage Probability

1: Input: $\boldsymbol{x}_i(t), \boldsymbol{y}_i(t), \boldsymbol{z}_i(t), \boldsymbol{i} \in [1, \boldsymbol{N}]$
2: Output: $P_O(t)$
3: Initialize $\boldsymbol{J} = [\boldsymbol{J}_1, \boldsymbol{J}_2, \ldots, \boldsymbol{J}_{2^N}]$ as the indicator matrix where $\boldsymbol{J}[k,:]$ is the kth combination and $\boldsymbol{J}[k,c] = 1$ if the link c is under LoS condition, and 0 otherwise
4: Initialize $\boldsymbol{M} = 0, \boldsymbol{\theta} = 0, \boldsymbol{\alpha} = 0, \boldsymbol{P}_{LoS} = 0, \boldsymbol{P}_{NLoS} = 0, \boldsymbol{D} = 0, \boldsymbol{M}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{P}_{LoS}, \boldsymbol{P}_{NLoS}, \boldsymbol{D} \in \mathbb{R}^{N \times 1}$
5: Initialize $\boldsymbol{P}_O = 0, \boldsymbol{P}_{res} = 1, \boldsymbol{P}_O, \boldsymbol{P}_{res} \in \mathbb{R}^{2^N \times 1}$
6: $\boldsymbol{D}[i] = \sqrt{(x_i(t) - x_0)^2 + (y_i(t) - y_0)^2}$, $\boldsymbol{P}_{LoS}[i] = P_{LoS}(z_i(t), \boldsymbol{D}[i])$ from (5), $\boldsymbol{P}_{NLoS}[i] = 1 - \boldsymbol{P}_{LoS}[i]$
7: for $k = 1 : 2^N$ do
8: for $c = 1 : N$ do
9: if $J_{k,c} == 1$ then
10: $\boldsymbol{M}[c] = m_{LoS}, \boldsymbol{\theta}[c] = \theta_{LoS}, \boldsymbol{\alpha}[c] = \alpha_{LoS}$
11: $\boldsymbol{P}_{res}[k] = \boldsymbol{P}_{res}[k] * \boldsymbol{P}_{LoS}[c]$
12: else
13: $\boldsymbol{M}[c] = m_{NLoS}, \boldsymbol{\theta}[c] = \theta_{NLoS}, \boldsymbol{\alpha}[c] = \alpha_{NLoS}$
14: $\boldsymbol{P}_{res}[k] = P_{res}[k] * \boldsymbol{P}_{NLoS}[c]$
15: end if
16: end for
17: $\boldsymbol{P}_O[k] = P_O(\gamma_{th}, \boldsymbol{M}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{D})$ from (20)
18: end for
19: return $P_o(t) = \boldsymbol{P}_{res}^T * \boldsymbol{P}_O$

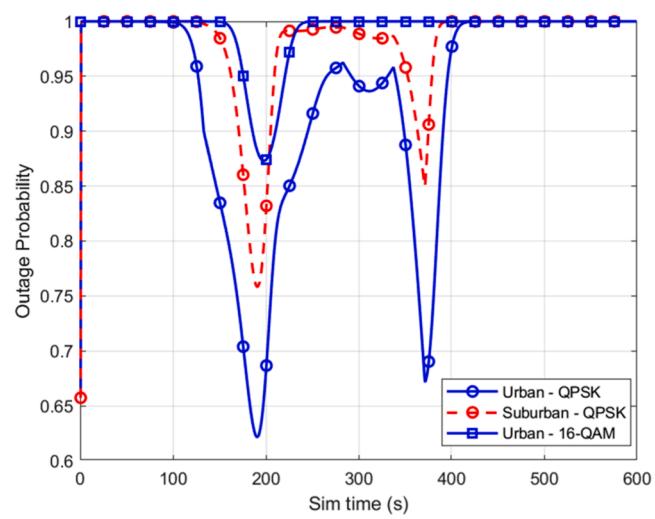

Fig. 5. Outage probability of the main link throughout the mission duration in urban and suburban environments. For the urban environment, two instances of the network with different Modulation and Coding Scheme are compared.

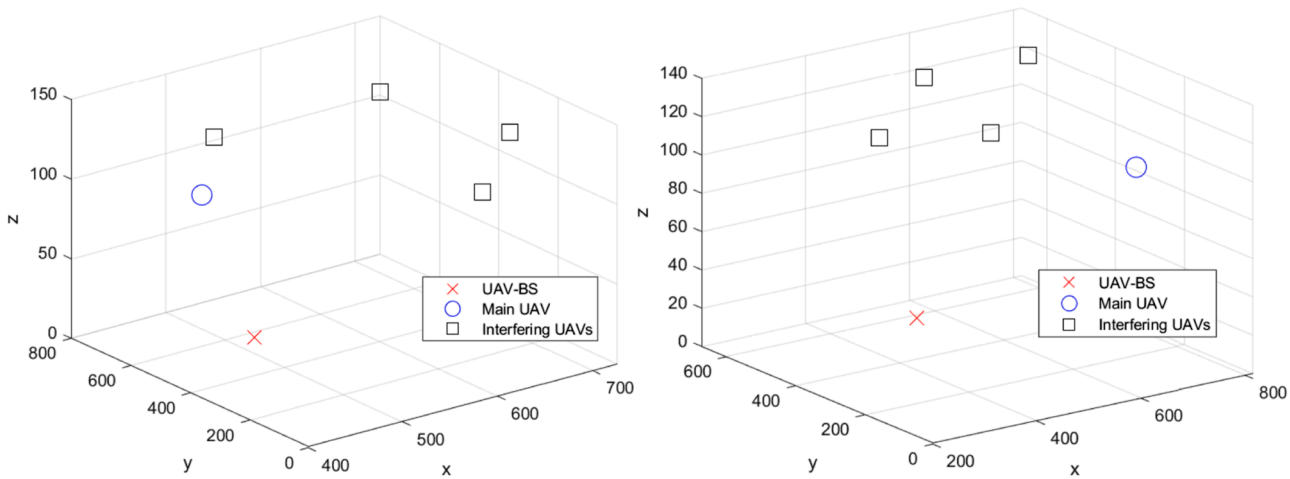

Fig. 6. Position of the UAVs during periods of low (left) and high (right) outage probabilities

SoI depends on the position of all the UAVs in the planned trajectory as their placements greatly impact the resultant SINR. When the main UAV, U_S, is the closest UAV to the UAV-BS, the received signal strength of the SoI at the UAV-BS is stronger than the interfering signals, leading to higher SINR and lower outage probability. Conversely, when the main UAV is far away from the UAV-BS, the magnitude of the aggregate interference from the interfering nodes is greater than the received signal strength of the SoI, which results in a high probability for an outage event to occur (relative to the threshold of 4dB).

Aside from exploring the baseline reliability of the network for different environments, the model can also be used to evaluate the reliabilities of different UAV communication systems such as one that is using a Quadrature Phase Shift Keying (QPSK) modulation scheme or a higher bitrate 16-Quadrature Amplitude Modulation (16-QAM) scheme. The main difference between these modulation schemes lies in the SINR threshold for successful reception of the signals as more sophisticated modulation schemes like the 16-QAM are less tolerant to noise and interference, hence a higher SINR threshold is required to ensure that the signals can be decoded correctly with a pre-defined Bit Error Rate (BER) [37]. This effect can be captured by using a higher SINR threshold while evaluating the outage probability using the proposed model. The difference in the temporal outage probability for these two instances of multi-UAV network in the urban environment with the same trajectory is also plotted in Fig. 5. For this example, the SINR threshold for QPSK is set as 4dB whereas the SINR threshold is 12dB for 16-QAM following from [37] to achieve BER of 10^{-3}. As seen from the plot, the communication system using QPSK is significantly more reliable than its counterpart using 16-QAM in the urban environment for the given trajectory although the bandwidth of the links can be theoretically increased with 16-QAM. This is because 16-QAM is less tolerant towards noises and interferences as it is a more sophisticated coding technique that has lower error margins. This implies that the QPSK modulation scheme is necessary to ensure reliable communication between the main UAV and the UAV-BS throughout the mission lifetime given the pre-planned UAV trajectories and the operating environment, even though this choice results in lower bandwidth.

With this information, various optimization efforts such as trajectory design, mission planning or proper tuning of the communication system parameters can be carried out to lower the outage probability of the FANETs or to maintain a minimum level of confidence on the reliability of the links necessary for successful mission completion. Actions that can be taken to potentially circumvent the high outage probability throughout the mission period include using less demanding MCS that will allow the receiver to tolerate lower SINR, using directional antennas so that the contribution of the interfering signals are reduced when the interfering nodes are not communicating with the UAV-BS or utilizing other higher-layer techniques such as using Code Division Multiple Access (CDMA) as a channel access scheme to minimize CCI. Therefore, the proposed model enables FANET operators to properly plan their multi-UAV deployments to ensure a baseline reliability is met in the network.

Another interesting analysis that can be carried out is to determine the expected outage duration of the network for a given mission using the proposed general Outage Probably model. This information can potentially be used as an optimization target to ensure that the multi-UAV network does not suffer from prolonged outage during the mission lifetime. This computation can be done by integrating the area under the outage probability curves for different environments or communication systems and is mathematically defined for the period t_0 to t_1 as:

$$E_t[Outage] = \int_{t_0}^{t_1} P_o(t)\ dt \tag{21}$$

To summarize, the case study in this subsection showcases how our proposed general outage probability model can be implemented and used to evaluate the network outage probability of a multi-UAV deployment in different environments and using different MCS given the trajectories. The omission of the Point Process assumption allows the model to be used in evaluating the temporal outage probability of the network which is previously lacking in the literature. Furthermore, we also demonstrated how the temporal outage probability can be further refined into an expected outage duration. This is an additional network metric that is tied closely to the environment, network design and trajectory planning. Therefore, it can be used as a metric for optimization by operators or researchers when planning for a multi-UAV mission.

V. Conclusion

Multi-UAV systems are imperative pieces of next generation Autonomous Mobile Robots (AMRs) that can complete complicated missions autonomously and collaboratively. Due to the complex environment that UAVs operate in and the dynamic mobilities of the UAVs, proper network and trajectory optimization must be performed prior to mission-based deployments to increase the mission success rate. However, most researchers resort to deterministic SINR models due to their simplicities whereas most stochastic SINR models are constrained by the Point Process or limiting network assumptions. In this paper, we addressed these two gaps by deriving and presenting an analytical closed-form mathematical model to characterize the CDF of the SINR of a main link by considering hybrid LoS channel and not making any assumptions about the node distribution. With the closed-form model, we

studied how the outage probability is affected by different channel parameters, specifically the path loss exponent and Nakagami fading shape parameter. We note that the outage probability of the network worsens as the degree of impairments in the channel increases. Our results also show that performance gain from better fading channels is only observed at low SINR threshold regimes. On top of that, we also showed that it is possible to analyze the channel capacity using our proposed model and the Shannon-Hartley Formula. Numerical results from evaluating the derived model match with the simulation results, hence affirming the robustness and accuracy of the proposed model in different network conditions. Furthermore, we have demonstrated one potential use-case of the proposed model by evaluating the outage probability of a main U2U link in a multi-UAV network using mobility trace data from OMNeT++ in two different environments. From the evaluations, we conclude that the multi-UAV network is more likely to experience outage events in the urban environment compared to the suburban environment, given the same mobility pattern, as the added attenuation and fading losses contribute to lower aggregate interference power. Furthermore, a multi-UAV network using a less sophisticated MCS such as QPSK is shown to have significantly better reliability than one using 16-QAM for the given UAV trajectories in the urban setting. These insights can then be used by operators to optimize the network reliability and performance by carefully choosing the right deployment or network parameters. Without making prior assumptions about the spatial distribution of the UAVs, we foresee that this model can be adopted by multi-UAV system operators or researchers to study the baseline Physical layer reliability of the network for mission-based deployments where the mobility and spacing of UAVs in the cluster are often predetermined. Furthermore, the closed-form model also provides operators with a means to optimize their multi-UAV deployments without oversimplifying the network assumption or resorting to complex optimization methods.

CRediT authorship contribution statement

Wei Jian Lau: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Validation, Writing – original draft, Writing – review & editing. **Joanne Mun-Yee Lim:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Chun Yong Chong:** Conceptualization, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Nee Shen Ho:** Project administration, Resources, Supervision. **Thomas Wei Min Ooi:** Project administration, Resources, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests

Data availability

Data will be made available on request.

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Appendix

A. Proof of Laplacian

The Laplacian $\mathscr{L}_I(s)$ is given by:

$$\mathscr{L}_I(s) = \int_0^\infty \exp(-sx) f_I(x) dx$$

$$= E_I\left[\prod_{u=1}^{N-1} \exp\left(-sgP_{L_u} h_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}\right)\right]$$

$$=^{(a)} \prod_{u=1}^{N-1} E_{h_{L_u}}\left[\exp\left(-sgP_{L_u} h_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}\right)\right] \tag{A.1}$$

where (a) results from the independence of the fading phenomenon between each interfering links. Expanding on the Expectation operator on (A.1) and letting z_{L_u} be drawn from the Gamma distribution with PDF, $f_{h_{L_u}}(Z_{L_u})$, reveals the following:

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} \int_0^\infty e^{-sgP_{L_u} h_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}} f_{h_{L_u}}(z_{L_u}) dz_{L_u} \tag{A.2}$$

Substituting the PDF expression of the Gamma Random Variable as given in (4) into (A.2) yields:

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} \int_0^\infty \frac{z_u^{m_{e_{L_u}}-1}}{\Gamma(m_{e_{L_u}})\theta_{e_{L_u}}^{m_{e_{L_u}}}} \exp\left(-z_{L_u}\left(\frac{1}{\theta_{e_{L_u}}} + sgP_u r_u^{-\alpha_{e_{L_u}}}\right)\right) dz_{L_u} \tag{A.3}$$

Let $\theta_{new} = \left(\frac{1}{\theta_{e_{L_u}}} + sgP_u r_u^{-\alpha_{e_{L_u}}}\right)^{-1}$, the above can be subsequently simplified to reveal:

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} \theta_{e_{L_u}}^{-m_{e_{L_u}}} \theta_{new}^{m_{e_{L_u}}} \int_0^{\infty} \frac{z_{L_u}^{m_{e_{L_u}}-1}}{\Gamma(m_{e_{L_u}})\theta_{new}^{m_{e_{L_u}}}} \exp\left(-\frac{z_u}{\theta_{new}}\right) dz_{L_u} \tag{A.4}$$

Since the integral in (A.4) evaluates to 1, (A.4) can be further simplified to reveal (A.5) which completes the proof.

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} \left(1 + sg\theta_{e_{L_u}} P_u r_u^{-\alpha_{e_{L_u}}}\right)^{-m_{e_{L_u}}} \tag{A.5}$$

B. Proof of Closed Form Expression

Notice from (17) that the Laplacian is given by the product of $(N-1)$ functions. Let $f_{L_u}(s) = (1 + sg\theta_{e_{L_u}} P_u r_u^{-\alpha_{e_{L_u}}})^{-m_{e_{L_u}}}$, we can rewrite the Laplacian as:

$$\mathscr{L}_I(s) = \prod_{u=1}^{N-1} f_{L_u}(s) \tag{B.1}$$

To apply the General Leibniz Rule, we find a general function to describe the nth derivative of the function $f_{L_u}(s)$. This can be derived by performing successive differentiation on $f_{L_u}(s)$. Assuming that $m_{e_{L_u}} \in Z^+$, the general expression of the nth derivative can be written as:

$$f_{L_u}^{(n)}(s) = \frac{(-1)^n (m_{e_{L_u}} + n - 1)!}{(m_{e_{L_u}} - 1)!} \times \left(1 + sg\theta_{e_{L_u}} P_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}\right)^{-m_{e_{L_u}} - n} \times \left(\theta_{e_{L_u}} g P_{L_u} r_{L_u}^{-\alpha_{e_{L_u}}}\right)^n \tag{B.2}$$

With a general function derived for the nth derivative of the function $f_{L_u}(s)$, we then apply the General Leibniz Rule to derive an expression for the nth derivative of the Laplacian, denoted by $H(s,n)$ as:

$$H(s,n) = \sum_{k_{L_1}+k_{L_2}+\ldots+k_{L_{N-1}}=n} \binom{n}{k_{L_1}, k_{L_2}, \ldots, k_{L_{N-1}}} \times \prod_{u \in \Psi \setminus U_S} f_{L_u}^{(k_u)}(s) \tag{B.3}$$

where $\binom{n}{k_{L_1}, k_{L_2}, \ldots, k_{L_{N-1}}}$ is the multinomial coefficient and the summation in (B.3) is performed over all possible $(N-1)$ tuples of non-negative integers and $L_1, L_2, \ldots, L_{N-1}$ are the indices of the interfering UAVs. Substituting the general closed-form expression in (B.3) back to (17) yields (19) which completes the proof.

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WEI JIAN LAU received the B.Eng. (Hons) in Electrical and Computer Systems Engineering from Monash University Malaysia in November 2020. He is currently a pursuing the Ph. D. degree in the same university in wireless communications on Unmanned Aerial Vehicles (UAV). His research interest includes UAV wireless communications, network intelligence, Artificial Intelligence/Machine Learning and IoT.

Joanne Mun-Yee Lim received her Ph.D. in Engineering from Multimedia University. She is also a member of The Institution of Engineers Malaysia and a Professional Engineer with Practicing Certificate (PEPC) of the Board of Engineers Malaysia. She joined the Department of Electrical Engineering in Asia Pacific University as a Lecturer in 2011. She is currently a Senior Lecturer in the Department of Electrical and Computer Systems Engineering, School of Engineering, Monash University Malaysia. Her research interest includes Internet of Things (IoT), Intelligent Transportation System (ITS), Unmanned Aerial Vehicle (UAV/Drones), Vehicular Ad-hoc Network (VANET), Artificial intelligence, Machine Learning, Optimization schemes, Robotics design and Sustainable Systems.

Chun Yong Chong (M'19) received the M.Sc. and Ph.D. degrees in computer science from the University of Malaya, Malaysia, in 2012 and 2016, respectively. He is currently a Senior Lecturer at the School of Information Technology, Monash University Malaysia.

His current research interests include software maintenance, software clustering, software remodularization, software fault prediction, and metamorphic testing. He was awarded Marie Skłodowska-Curie Research Exchange Fellowship and visited XLAB d.o.o., Slovenia, as a visiting scholar under the EU Project IDENTITY 690907.

Nee Shen Ho received his M.Sc in Reliable Embedded Systems from University of Leicester in 2013 and Bachelor's degree from Multimedia University in 2004. He is currently a Software Program Manager at Intel Research and Development Ireland working on Intel's next generation artificial intelligence accelerator. He used to coordinate strategic university program engagement for Intel's Malaysia Design Center.

Dr Thomas Ooi, currently serving as Director of Central Solutions Enabling in Intel Network & Edge Solutions Group, has >20 years of practical experience in business management, technical services, telecommunication and data communication engineering, software product development and deployment, and program management etc in both services and product industry. Currently he also served as the APJ leader for Federal and Industrial business for Intel Network and Edge Solutions Group apart from his global role as the Director of Central Solutions Scaling. Thomas holds a bachelor degree in Computer Science, MBA and Doctorate in Business Management.